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17.

ELEMENTARY RESEARCHES IN THE ANALYSIS OF COMBINATORIAL AGGREGATION

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The ensuing inquiries will be found to relate to combination-systems, that is, to combinations viewed in an aggregative capacity, whose species being given, we shall have to discover rules for ranging or evolving them in classes amenable to certain prescribed conditions. The question of numerical amount will only appear incidentally, and never be made the primary object of investigation¹.

The number of things combined will be termed the modulus of the system to which they belong. The elements taken singly, or combined in twos, threes, &c., will be denominated accordingly the monadic, duadic, triadic elements, or simply the monads, duads, or triads of the system.

Let us agree to denote by the word *synthème*² any aggregate of combinations in which all the monads of a given system appear once, and once only.

It is manifest that many such *synthèmes* totally diverse in every term may be obtained for a given system to any modulus, and for any order of combination.

Let us begin with considering the case of duad *synthèmes*. Take the modulus 4 and call the elements a, b, c, d .

$$(ab, cd), \quad (ac, bd), \quad (ad, cb)$$

constitute three perfectly independent *synthèmes*, and these three *synthèmes* include between them all the duad elements, so that no more independent *synthèmes* can be obtained from them.

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Again, let a, b, c, d, e, f be the monads; we can write down five independent *synthèmes*, to wit,

$$\left. \begin{array}{lll} ab, & cd, & ef \\ ad, & cf, & eb \\ ac, & de, & fb \\ af, & bd, & ce \\ ae, & df, & bc \end{array} \right\}.$$

We can write no more than these without repeating duads which have already appeared³.

¹The present theory may be considered as belonging to a part of mathematics which bears to the combinatorial analysis much the same relation as the geometry of position to that of measure, or the theory of numbers to computative arithmetic; number, place, and combination (as it seems to the author of this paper) being the three intersecting but distinct spheres of thought to which all mathematical ideas admit of being referred.

²From $\sigma\nu\nu$ and $\tau\iota\theta\eta\mu\iota$.

³Such an aggregate of *synthèmes* may be therefore termed a Total.

We propose to ourselves this problem:—A system to any even⁴ modulus being given, to arrange the whole of its duads⁵ in the form of syntheses; or in other words, to evolve a Total of duad syntheses to any given even modulus⁶.

When the modulus is odd, as before remarked, the formation of a duad synthesis is of course impossible, for any number of duads must necessarily contain an even number of monadic elements; but there is nothing to prevent us from forming in all cases what may be termed a bisynthesis or diplotheism, that is, an aggregate of combinations, where each element occurs twice and no more.

For instance, if the elements be called after the letters of the alphabet, we have

$$\left(\begin{array}{l} ab, bc, cd, de, ea \\ ac, ce, eb, bd, da \end{array} \right),$$

the bisynthetic total to modulus 5; and in like manner

$$\left. \begin{array}{l} ab, bc, cd, de, ef, fg, ga \\ ac, ce, eg, gb, bd, df, fa \\ ad, dg, gc, cf, fb, be, ea \end{array} \right\} \text{ the total to modulus 7.}$$

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In general, if n be the modulus, the number of duads is $n \frac{n-1}{2}$; n being even, $\frac{n}{2}$ duads go to each synthesis, and therefore the total contains $(n-1)$ of these.

⁴The modulus must be even, as otherwise it is manifest no single synthesis can be formed. We shall before long extend the scope of our inquiry so as to take in the case of odd moduli.

⁵Triadic systems will be treated of hereafter.

⁶It is scarcely necessary to advert here to the fact of the problem being in general indeterminate and admitting of a great variety of solutions.

When the modulus is four there is only one synthetic arrangement possible, and there is no indeterminateness of any kind; from this we can infer, *a priori*, the reducibility of a biquadratic equation; for using Φ, ϕ, F to denote rational symmetrical forms of function, it follows that

$$F \left\{ \begin{array}{l} f\{\phi(a,b), \phi(c,d)\} \\ f\{\phi(a,c), \phi(b,d)\} \\ f\{\phi(a,d), \phi(b,c)\} \end{array} \right\}$$

is itself a rational symmetric function of a, b, c, d . Whence it follows that if a, b, c, d be the roots of a biquadratic equation, $f\{\phi(a,b), \phi(c,d)\}$ can be found by the solution of a cubic: for instance, $(a+b) \times (c+d)$ can be thus determined, whence immediately the sum of any two of the roots comes out from a quadratic equation.

To the modulus 6 there are fifteen different syntheses capable of being constructed; at first sight it might be supposed that these could be classed in natural families of three or of five each, on which supposition the equation of the sixth degree could be depressed; but on inquiry this hope will prove to be futile, not but what natural affinities do exist between the totals; but in order to separate them into families each will have to be taken twice over, or in other words, the fifteen syntheses to modulus 6 being reduplicated subdivide into six natural families of five each. Again, it is true that the triads to modulus 6 (just like the duads to modulus 4) admit of being thrown into but one synthetic total, but then this will contain ten syntheses, a number greater than the modulus itself.

If n be odd, then, since always n duads go to a bisyntheme, the number of such in the total is $\frac{n-1}{2}$.

Before proceeding to the solution of the problem first proposed, let us investigate the theory of diplothematic arrangement. Here we shall find another term convenient to employ. By a cyclotheme, I designate a fixed arrangement of the elements in one or more circles, in which, although for typographical purposes they are written out in a straight line, the last term is to be viewed as contiguous and antecedent to the first; the recurrence may be denoted by laying a dot upon the two opened ends of the circle; $\dot{a}.b.c.d.\dot{e}$ will thus denote a cyclotheme to modulus 5; $\dot{a}.b.c.d.e.f.g.h.\dot{k}$ the same to modulus 9; so also is $\dot{a}.b.c, \dot{d}.e.f, \dot{g}.h.k$ a cyclotheme of another species to the same modulus. In general the number of terms will be alike in each division of a cyclotheme.

Now it is evident that every cyclotheme, on taking together the elements that lie in conjunction, may be developed into a diplotheme. Thus

$$\begin{aligned}\dot{1}.2.\dot{3} &= 12, 23, 31, \\ \dot{1}.2.3.\dot{4} &= 12, 23, 34, 41, \\ (\dot{1}.2.\dot{3}; \dot{4}.5.\dot{6}; \dot{7}.8.\dot{9}) &= \begin{pmatrix} 12, & 23, & 31 \\ 45, & 56, & 64 \\ 78, & 89, & 97 \end{pmatrix}.\end{aligned}$$

Hence we shall derive a rule for throwing the duads of any system into bisynthemes.

Let $m = 3$, we have simply $\dot{a}\dot{b}\dot{c}$;

$$m = 5, \text{ we write } \begin{array}{l} \dot{a}.b.c.d.\dot{e}, \\ \dot{a}.c.e.b.\dot{d}, \end{array}$$

the second being derived from the first by omitting every alternate term; similarly below, the lines are derived each from its antecedent.

$$m = 7, \text{ we have } \begin{array}{l} \dot{a}.b.c.d.e.f.\dot{g}, \\ \dot{a}.c.e.g.b.d.\dot{f}, \\ \dot{a}.e.b.f.c.g.\dot{d}. \end{array}$$

A very little consideration will serve to prove that in this way, m being a prime number, $\frac{m-1}{2}$ cyclothemes may be formed, such that no element will ever be found more than once in contact on either side with any other; whence the rule for obtaining the diplothematic total to any prime-number modulus is apparent.

For example, to modulus 7 the total reads thus:—

$$\begin{array}{lllllll} \text{1st.} & ab, & bc, & cd, & de, & ef, & fg, & ga \\ \text{2nd.} & ac, & ce, & eg, & gb, & bd, & df, & fa \\ \text{3rd.} & ae, & eb, & bf, & fc, & cg, & gd, & da \end{array}$$

and no more remains to be said on this special case.

Let us now return to the theory of even moduli, and show how to apply what has been just done to constructing a synthematic total to a modulus which is the double of a prime number.

Suppose the modulus to be six, the number of syntheses is five. Let the six elements, a, b, c, d, e, f , be taken in three parts, so that each part contains two of them; let these parts be called A, B, C , where A denotes ab , B, cd , and C, ef .

Now the duads will evidently admit of a distinction into two classes, those that lie in one part, and those that lie between two; thus ab, cd, ef will be each unipartite duads, the rest will be bipartite.

The unipartite duads may be conveniently formed into a syntheme by themselves; it only remains to form the four remaining bipartite duad syntheses.

Write the parts in cyclothematic order, as below:

$$\dot{A}\dot{B}\dot{C}.$$

It will be observed that each part may be written in two positions; thus

$$\begin{array}{rcll} A & \text{may be expressed by} & \begin{array}{c} a \\ b \end{array} & \text{or by} \quad \begin{array}{c} b \\ a' \end{array} \\ B & & \begin{array}{c} c \\ d \end{array} & \begin{array}{c} d \\ c' \end{array} \\ C & & \begin{array}{c} e \\ f \end{array} & \begin{array}{c} f \\ e' \end{array} \end{array}$$

Now we may form a cyclic table of positions as below:

$$\begin{array}{ccc} \dot{A} & B & \dot{C} \\ \hline 1 & 1 & 1 \\ 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{array}$$

Here the numbers in each horizontal line denote the synchronic positions of the parts. p. 95

On inspection it will be discovered that A will be found in each of its two positions, with B in each of its two; similarly B with C , and C with A . In fact the four permutations, 11, 12, 21, 22, occur, though in different orders, in any two assigned vertical columns.

Now develop the preceding table, and we have

$$\begin{array}{cccc} ace & adf & bcf & bde, \\ bdf & bce & ade & acf; \end{array}$$

and these being read off (the *superior* of each antecedent with the *inferior* of each consequent⁷) must manifestly give the four independent bipartite syntheses which we were in quest of, *videlicet*

$$(ad, cf, eb), \quad (ac, de, fb), \quad (bd, ce, fa), \quad (bc, df, ea) :$$

these four, together with the syntheme first described (ab, cd, ef), constitute a duad synthemetic total to modulus 6.

Before proceeding further let us take occasion to remark that the foregoing table of positions may evidently be extended to any odd number of terms by repetition of the second and third places, as seen in the annexed tables of position.

$$\begin{array}{ccccccccc} \dot{1} & 1 & 1 & \dot{1} & \dot{1} & 1 & 1 & 1 & 1 & \dot{1} \\ \dot{1} & 2 & 2 & \dot{2} & \dot{1} & 2 & 2 & 2 & 2 & \dot{2} \\ \dot{2} & 1 & 2 & \dot{1} & \dot{2} & 1 & 2 & 1 & 2 & \dot{2} \\ \dot{2} & 2 & 1 & \dot{1} & \dot{2} & 2 & 1 & 2 & 1 & \dot{2} \end{array}$$

Now let 10 be the modulus.

As before divide the elements into five parts, which call A, B, C, D, E .

The unipartite duads fall into a single syntheme; the eight remaining bipartite syntheses may be found as follows:—

Arrange in cyclothemes $\left(\frac{n-1}{2} \text{ in number}\right)$ the odd modulus system A, B, C, D, E .

We have thus

$$\dot{A}BCD\dot{E}, \quad \dot{A}CEB\dot{D}.$$

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Let each cyclotheme be taken in the four positions given in the table above, we have thus 2×4 , that is, eight arguments.

$$\begin{array}{cccc} \dot{a}bcd\dot{e}, & \dot{a}\beta\gamma\delta\dot{\epsilon}, & \dot{a}b\gamma d\dot{e}, & \dot{a}\beta c\delta\dot{e}, \\ \alpha\beta\gamma\delta\epsilon, & abcde, & a\beta cde, & ab\gamma de, \\ \dot{a}cebd\dot{1}, & \dot{a}\gamma e\beta\dot{\delta}, & \dot{a}ceb\dot{\delta}, & \dot{a}\gamma e\beta\dot{d}, \\ \alpha\gamma e\beta\delta, & acebd, & a\gamma e\beta d, & acebd. \end{array}$$

And each of these arguments will furnish one bipartite syntheme, by reading off, as before, the superior of each antecedent with the inferior of each consequent; and the least reflection will serve to show that the same duad can never appear in two distinct arguments.

In like manner, if the modulus be 14 and seven parts be taken, the bipartite syntheses, twelve in number, may be expressed symbolically thus:

$$\left(\begin{array}{c} \dot{1}.1.1.1.1.1.\dot{1} \\ +\dot{1}.2.2.2.2.2.\dot{2} \\ +\dot{2}.1.2.1.2.1.\dot{2} \\ +\dot{2}.2.1.2.1.2.\dot{1} \end{array} \right) \times \left(\begin{array}{c} \dot{A}.B.C.D.E.F.\dot{G} \\ +\dot{A}.C.E.G.B.D.\dot{F} \\ +\dot{A}.E.B.F.C.G.\dot{D} \end{array} \right).$$

⁷Any other fixed order of successive conjunction would answer equally well.

Nay more, from the above table, if we agree to name the elements

$$\begin{array}{cc} A_1 & B_1 \\ A_2 & B_2 \end{array}, \quad \&c.,$$

we can at once proceed to calculate each of the twelve synthemes in question by an easy algorithm. For instance,

$$(\dot{1}.2.2.2.2.\dot{2}) \times (\dot{A}.C.E.G.B.D.\dot{F}) = (A_1C_1, C_2E_1, E_2G_1, G_2B_1, B_2D_1, D_2F_1, F_2A_2).$$

And again

$$(\dot{2}.1.2.1.2.1.\dot{2}) \times (\dot{A}.E.B.F.C.G.\dot{D}) = A_2E_2, E_1B_1, B_2F_2, F_1C_1, C_2G_2, G_1D_1, D_2A_1;$$

each figure occurring once unchanged as an antecedent and once changed as a consequent.

If it were thought worth while it would not be difficult, by using numbers instead of letters, to obtain a general analytical formula, from which all similarly constituted synthemes to any modulus might be evolved.

But the rule of proceeding must be now sufficiently obvious; the modulus being $2p$, we divide the elements into p classes; these may be arranged into $\frac{p-1}{2}$ distinct forms of cyclothematic arrangement, and each of the cyclothemes taken in four positions, thus giving $4 \times \frac{p-1}{2}$, that is, $2p-2$ bipartite synthemes, the whole number that can be formed to the given modulus $2p$.

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I shall now proceed to the theory of bipartite synthemes to the modulus $2m \times p$, by which it is to be understood that we have p parts each containing $2m$ terms, and p is at present supposed to be a prime number; the total number of synthemes to the modulus $2mp$ being $2mp-1$, and $2m-1$ of these evidently being capable of being made unipartite; the remainder, $2mp-2m$, that is, $(p-1)2m$, will be the number of bipartites to be obtained⁸:

$$2m(p-1) = \frac{p-1}{2} \times 4m;$$

$\frac{p-1}{2}$ denotes the total number of cyclothemes to modulus p ; $4m$, as will be presently shown, the number of lines or syzygies in the Table of position.

To fix our ideas let the modulus be 4×3 , and let A, B, C be three parts:

$$\left. \begin{array}{cccc} a_1 & a_2 & a_3 & a_4, \\ b_1 & b_2 & b_3 & b_4, \\ c_1 & c_2 & c_3 & c_4 \end{array} \right\} \text{ their constituents respectively.}$$

⁸In general, if there be π parts of μ terms each, and $\mu\pi$ be even, the number of bipartite synthemes is $(\pi-1)\mu$, as is easily shown from dividing the whole number of bipartite duads by the semi-modulus.

Give a fixed order to the constituents of each part, then each of them may be taken in four positions; thus A may be written

$$\begin{array}{cccc} a_1 & a_2 & a_3 & a_4, \\ a_2 & a_3 & a_4 & a_1, \\ a_3 & a_4 & a_1 & a_2, \\ a_4 & a_1 & a_2 & a_3. \end{array}$$

Assume some particular position for each, as, for instance,

$$\begin{array}{ccc} a_1 & b_1 & c_1, \\ a_2 & b_2 & c_2, \\ a_3 & b_3 & c_3, \\ a_4 & b_4 & c_4, \end{array}$$

and read off by coupling the first and third vertical places of each antecedent with the second and fourth respectively of each consequent; we have accordingly,

$$a_1b_2, b_1c_2, c_1a_2, \quad a_3b_4, b_3c_4, c_3a_4.$$

It is apparent that the same combinations will recur if any two contiguous parts revolve simultaneously through two steps; or in other words, that $A_rB_s = A_{r+2}B_{s+2}$, where μ is any number, odd or even.

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Symbolically speaking, therefore, as regards our table of position,

$$r : s = r + 2 : s + 2,$$

or more generally,

$$= r + 2 \pm 4i : s + 2 \pm 4i.$$

So that

$$\begin{array}{ll} 1 : 1 = 3 : 3, & 2 : 1 = 4 : 3, \\ 1 : 2 = 3 : 4, & 2 : 2 = 4 : 4, \\ 1 : 3 = 3 : 1, & 2 : 3 = 4 : 1, \\ 1 : 4 = 3 : 2, & 2 : 4 = 4 : 2. \end{array}$$

There are therefore no more than eight independent unequivalent permutations to every pair of parts. Now inspect the following table of position:—

$$\begin{array}{ll} 1.1.1, & 2.1.2, \\ 1.2.3, & 2.2.4, \\ 1.3.2, & 2.3.1, \\ 1.4.4, & 2.4.3. \end{array}$$

It will be seen that in the first and second, second and third, third and first places, all the eight independent permutations occur under different names; the law of formation of such and similar tables will be explained in due time; enough

for our present object to see how, by means of this table, we are able to obtain the bipartite syntheses to the given modulus 4×3 ; the number according to our formula is $2 \times 4 \times \frac{3-1}{2} = 8$, and they may be denoted symbolically as follows:—

$$(\dot{A}.B.\dot{C}) \begin{pmatrix} 1.1.1 + 1.2.3 + 1.3.2 + 1.4.4 \\ +2.1.2 + 2.2.4 + 2.3.1 + 2.4.3 \end{pmatrix}.$$

Each of the eight terms connected by the sign of + gives a distinct syntheme; for example, let us operate on

$$\dot{A}.B.\dot{C} \times (2.3.1).$$

2.3.1 denotes 2.3, 3.1, 1.2.

$$2.3 \text{ gives rise to } 2(3+1) + (2+2).(3+3) = 2.4 + 4.2.$$

$$3.1 \text{ gives rise to } 3(1+1) + (3+2).(1+3) = 3.2 + 1.4.$$

$$1.2 \text{ gives rise to } 1(2+1) + (1+2).(2+3) = 1.3 + 3.1.$$

The syntheme in question is therefore

$$A_2B_4, A_4B_2, B_3C_2, B_1C_4, C_1A_3, C_3A_1,$$

and so on for all the rest, the rule being that

$$r : s = r(s+1) + (r+2)(s+3).$$

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Now, as before, it is evident that if we look only to contiguous terms, the above table of position may be extended to any number of odd terms, simply by repetition of the second and third figures in each syzygy; and hence the rule for obtaining the bipartite syntheses to the modulus $4 \times p$ is apparent. For instance, let $p = 7$, there will be $8 \times \frac{7-1}{2}$, that is, 8×3 of them denoted as follows:—

$$\left\{ \begin{array}{l} \dot{A}.B.C.D.E.F.\dot{G} \\ +\dot{A}.C.E.G.B.D.\dot{F} \\ +\dot{A}.E.B.F.C.G.\dot{D} \end{array} \right\} \times \left\{ \begin{array}{l} 1.1.1.1.1.1.1 + 2.1.2.1.2.1.2 \\ +1.2.3.2.3.2.3 + 2.2.4.2.4.2.4 \\ +1.3.2.3.2.3.2 + 2.3.1.3.1.3.1 \\ +1.4.4.4.4.4.4 + 2.4.3.4.3.4.3 \end{array} \right\}.$$

As an example of the mode of development, let us take the term

$$\dot{A}.E.B.F.C.G.\dot{D} \times \dot{2}.4.3.4.3.4.\dot{3},$$

$$\begin{aligned} \dot{2}.4.3.4.3.4.\dot{3} &= (2 : 4, 4 : 3, 3 : 4, 4 : 3, 3 : 4, 4 : 3, 3 : 2) \\ &= \begin{pmatrix} 2.1 & 4.4 & 3.1 & 4.4 & 3.1 & 4.4 & 3.3 \\ +4.3 & +2.2 & +1.3 & +2.2 & +1.3 & +2.2 & +1.1 \end{pmatrix}, \end{aligned}$$

$$\dot{A}.E.B.F.C.G.\dot{D} = AE, EB, BF, FC, CG, GD, DA,$$

and the product

$$= \begin{pmatrix} A_2E_1, & E_4B_4, & B_3F_1, & F_4C_4, & C_3G_1, & G_4D_4, & D_3A_3 \\ A_4E_3, & E_2B_2, & B_1F_3, & F_2C_2, & C_1G_3, & G_2D_2, & D_1A_1 \end{pmatrix}.$$

Let the modulus be 6×3 ; as before, give a fixed cyclic order to the constituents of each part, and each will admit of being exhibited in six positions.

Write similarly as before,

$$\begin{array}{lll} a_1 & b_1 & c_1, \\ a_2 & b_2 & c_2, \\ a_3 & b_3 & c_3, \\ a_4 & b_4 & c_4, \\ a_5 & b_5 & c_5, \\ a_6 & b_6 & c_6, \end{array}$$

and take the odd places of each antecedent with the even places of each consequent; it will now be seen that

$$r : s = r + 2 : s + 2 = r + 4 : s + 4,$$

and the number of independent permutations is $\frac{6.6}{3} = 2.6$; and so in general, if there be $2m$ constituents in a part, the number of independent permutations is

$$\frac{2m.2m}{m} = 4m.$$

The rule for the formation of the table will be apparent on inspection. I suppose only three parts, as the rule may always be extended to any number by reiteration of the second and third terms. The table will be found to resolve itself naturally into four parts, each containing m lines.

Let $m = 1$, we have

$$\begin{array}{ll} 1.1.1 & 2.1.2 \\ 1.2.2 & 2.2.1 \end{array}$$

$m = 2$, we have

$$\begin{array}{ll} 1.1.1 & 2.1.2 \\ 1.2.3 & 2.2.4 \\ 1.3.2 & 2.3.1 \\ 1.4.4 & 2.4.3 \end{array}$$

$m = 3$, we have

$$\begin{array}{ll} 1.1.1 & 2.1.2 \\ 1.2.3 & 2.2.4 \\ 1.3.5 & 2.3.6 \\ 1.4.2 & 2.4.1 \\ 1.5.4 & 2.5.3 \\ 1.6.6 & 2.6.5 \end{array}$$

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$m = 4$, we have

1.1.1	2.1.2
1.2.3	2.2.4
1.3.5	2.3.6
1.4.7	2.4.8
1.5.2	2.5.1
1.6.4	2.6.3
1.7.6	2.7.5
1.8.8	2.8.7

So that x , going through all its values from 1 to m , the general expression for the four parts is

$$\Sigma \left\{ \begin{array}{l} 1.x(2x-1) + 1(m+x)2x \\ +2.x.2x + 2(m+x)(2x-1) \end{array} \right\}.$$

To show the use of this formula, let us suppose that we have seven parts, each containing ten terms, the general expression for the bipartite duad synthemes is

$$\left\{ \begin{array}{l} \dot{A}.B.C.D.E.F.\dot{G} \\ +\dot{A}.C.E.G.B.D.\dot{F} \\ +\dot{A}.E.B.F.C.G.\dot{D} \end{array} \right\} \times \Sigma \left\{ \begin{array}{l} 1.x(2x-1)x(2x-1)x(2x-1) \\ +2.x.2x.x.2x.x.2x \\ +1(5+x)2x(5+x)2x(5+x)2x \\ +2(5+x)(2x-1)(5+x)(2x-1)(5+x)(2x-1) \end{array} \right\}.$$

Make, for example, $x = 3$, one of the synthemes in question out of the twelve corresponding to this value will be

$$\dot{A}.C.E.G.B.D.\dot{F} \times \dot{2}.3.6.3.6.3.\dot{6}.$$

Here

$$\dot{A}.C.E.G.B.D.\dot{F} = AC, CE, EG, GB, BD, DF, FA,$$

$$\dot{2}.3.6.3.6.3.\dot{6} = \begin{pmatrix} 2.4 & 3.7 & 6.4 & 3.7 & 6.4 & 3.7 & 6.3 \\ +4.6 & +5.9 & +8.6 & +5.9 & +8.6 & +5.9 & +8.5 \\ +6.8 & +7.1 & +10.8 & +7.1 & +10.8 & +7.1 & +10.7 \\ +8.10 & +9.3 & +2.10 & +9.3 & +2.10 & +9.3 & +2.9 \\ +10.2 & +1.5 & +4.2 & +1.5 & +4.2 & +1.5 & +4.1 \end{pmatrix},$$

and the product

$$\begin{array}{ccccccc} = A_2C_4, & C_3E_7, & E_6G_4, & G_3B_7, & B_6D_4, & D_3F_7, & F_6A_3 \\ A_4C_6, & C_5E_9, & E_8G_6, & G_5B_9, & B_8D_6, & D_5F_9, & F_8A_5, \\ & \&c. & & \&c. & & \&c. \end{array}$$

To prove the rule for the table of formation, it will be sufficient to show that no two contiguous duads ever contain the same or equivalent permutations; the equation of equivalence it will be remembered is

$$r : s = r + 2i \pm 2m : s + 2i \pm 2m.$$

Now, as regards the first and second terms, it is manifest that $1 : x$ cannot be equivalent, either to $1 : x'$ nor to $2 : x$, nor to $2 : x'$, where x' is any number differing from x .

Similarly, as regards the last and first terms, $x : 1$ cannot be equivalent to $x' : 1$, nor to $x : 2$, nor to $x' : 2$; therefore there is no danger as far as the first term is concerned, either as antecedent or consequent.

Again, it is clear that $x : (2x - 1)$ cannot interfere with $x' : 2x'$, nor $(m + x) : 2x$ with $(m + x') : (2x' - 1)$; neither can $(2x - 1) : x$ with $2x' : x'$, nor $2x : (m + x)$ with $(2x' - 1) : (m + x')$.

Again, if possible, let

$$x : (2x - 1) = (m + x') : (2x' - 1);$$

then

$$m + x' - x = 2i,$$

and

$$2x' - 2x = 2i,$$

therefore

$$2m = 2i,$$

or

$$m = i,$$

which is impossible, since $\pm i$ is the difference between two indices, each less than m .

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Similarly,

$$m + x : 2x \text{ cannot } = x' : 2x',$$

and *vice versa* with the terms changed

$$2x : (m + x) \text{ cannot } = 2x' : x',$$

and

$$(2x - 1) : x \text{ cannot } = (2x' - 1) : (m + x'),$$

which proves the rule for the table of formation.

So much for the bipartite duad synthemes. As regards the unipartite synthemes little need be said, for every part may be treated as a separate system, and as each will produce an equal number of synthemes, these being taken one with another, will furnish just as many unipartite synthemes of the whole system as there are synthemes due to each part. Thus then the synthemetic resolution of the modulus $2m \times p$ may be made to depend on the synthemization of $2m$ and the cyclothematization of p . This has been already shown (whatever m may be) for the case of p being a prime number; but I proceed now to extend the rule to the more general case of p being any number whatever.

18.

ON THE EXISTENCE OF ABSOLUTE CRITERIA FOR DETERMINING THE ROOTS OF NUMERICAL EQUATIONS

[*Philosophical Magazine*, XXV. (1844), pp. 442-445]

p. 103

I wish to indicate in this brief notice a fact which I believe has escaped observation hitherto, that there exist, certainly in some cases, and probably in all, infallible criteria for determining whether a given equation has all its roots rational or not.

In the equation of the second degree it is enough, in order that this may be the case, that the expression for the square of the difference of the roots shall be a perfect square; in other words, if $x^2 - px + q = 0$ have its roots rational, $p^2 - 4q$ must be not only a positive number (the condition of the roots being real), but that number must also be a complete square. In this case it is further evident that p must be either prime to q , or if not, the greatest common measure of p^2 and q must be a perfect square; but this condition is contained in the former, which is a sufficient criterion in itself.

If we now consider the equation of the third degree,

$$x^3 - px^2 + qx - r = 0,$$

one condition is, that the product of the squared differences shall be a perfect square; in other words, the equation cannot have all its roots rational unless

$$p^2q^2 - 4q^3 - 18pqr - 4p^3r - 27r^2$$

be a positive square number.

This remark is made at the end of the second supplement of Legendre's *Theory of Numbers*, and is indeed self-evident; and in like manner one condition may be obtained for an equation of any degree which is to have all its roots rational; but this is far from being the sole condition required.

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In the equation of the third degree, however, one other condition, conjoined with that above expressed, will serve to determine infallibly whether all the roots are rational or not.

To obtain this condition, let us suppose that by making $3x = y + p$ we obtain the equation

$$y^3 - Qy - R = 0.$$

Calling the three roots of this new equation α, β, γ (all of which it is evident must be rational if those of the first equation are so), we have

$$\alpha + \beta + \gamma = 0,$$

$$Q = -(\alpha\beta + \alpha\gamma + \beta\gamma) = \alpha^2 + \alpha\beta + \beta^2,$$

$$R = \alpha\beta\gamma.$$

From the last two equations it is easily seen that if k be any prime factor common to Q and R , k^2 will be contained in Q , and k^3 in R ; or, in other words, k will be a common measure of α, β, γ .

We have therefore a *second condition*, that $9q - 3p^2$ shall be a negative quantity, which is either prime to $2p^3 - 9qp + 27r$, or else so related to it, that the greatest common measure of the cube of the first and the square of the second is a perfect sixth power.

I now proceed to show the converse, that if these two conditions be both satisfied (and it will appear in the course of the inquiry that the first does *not* involve the second), the roots cannot help being all rational.

It is evident that the two conditions in question are tantamount to supposing that the roots of the proposed equation are linearly connected with those of another $z^3 - Qz - R = 0$ (by virtue of the assumption $3x = kz + p$), where Q may be considered as prime to R ; and where $4Q^3 - 27R^2$ is a perfect square.

Let now $4Q^3 - 27R^2 = D^2$, then $D^2 + 27R^2 = 4Q^3$, or $D^2 + 3(3R)^2 = 4Q^3$.

Here, as Q is prime to R , D can have no common measure but 3, with $3R$.

Firstly, let Q be prime to $3R$.

Then putting $f^2 + 3g^2 = Q^3$, the complete solution of the equation immediately preceding is contained in the two systems:

$$\begin{aligned} \text{1st. } D &= 2f, & 3R &= 2g. \\ \text{2nd. } D &= (f \pm 3g), & 3R &= f \mp g, \end{aligned}$$

and for both systems,

$$f \pm g\sqrt{-3} = \{h \pm 3k\sqrt{-3}\}^3.$$

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The second system must therefore be rejected, for g evidently contains 3, and therefore $f = 3R \pm g$ will contain 3, and therefore D and therefore Q will do the same, contrary to supposition.

Hence

$$\begin{aligned} \sqrt[3]{\left[\frac{R}{2} \pm \sqrt{\left\{-\left(\frac{Q^3}{27} - \frac{R^2}{4}\right)\right\}}\right]} &= \sqrt[3]{\left\{\frac{R}{2} \pm \frac{D}{2} \sqrt{\left(-\frac{1}{27}\right)}\right\}} \\ &= \sqrt[3]{\left\{\frac{g}{3} \pm f \sqrt{\left(-\frac{1}{27}\right)}\right\}} \\ &= \mp \frac{1}{3\sqrt{-3}} \sqrt[3]{\{f \pm g\sqrt{-3}\}} \\ &= -K \pm \frac{h}{3\sqrt{-3}} = \lambda \pm \mu\sqrt{-3}; \end{aligned}$$

and the three roots of the equation being

$$\begin{cases} \{\lambda + \mu\sqrt{-3}\} + \{\lambda - \mu\sqrt{-3}\}, \\ \frac{1 \pm \sqrt{-3}}{2} \{\lambda + \mu\sqrt{-3}\} + \frac{1 \mp \sqrt{-3}}{2} \{\lambda - \mu\sqrt{-3}\}, \end{cases}$$

will evidently be all rational, which of course includes the necessity of their being also integer.

Again, secondly, if we suppose that Q does contain 3, D^2 will contain 27, and consequently D will contain 9; and we shall have

$$R^2 + 3 \left(\frac{D}{9} \right)^2 = 4 \left(\frac{Q}{3} \right)^3.$$

Here R being prime to $\frac{D}{9}$, it may be shown, as in the last case, that the complete solution is

$$\frac{R}{2} \pm \frac{D}{18} \sqrt{-3} = \{h \pm k\sqrt{-3}\}^3;$$

consequently

$$\sqrt[3]{\left\{ \frac{R}{2} \pm \sqrt{\left(\frac{R^2}{4} - \frac{Q^3}{27} \right)} \right\}} = h \pm k\sqrt{-3};$$

and the three roots of the equation are

$$2h, \quad h - 3k, \quad h + 3k$$

respectively, and are therefore all rational.

Here it may be observed that the condition of R being an even number, which we know, *à priori*, is the case when all the roots are rational, is

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involved in the two more general conditions already expressed. It will now be evident that the first condition by no means involves the second, as it is perfectly easy to satisfy the equation $f^2 + 3g^2 = Q^3$ without supposing anything relative to k , the common measure of f, g, Q , except that it be itself of the form $\lambda^2 + 3\mu^2$, which will give

$$\left(\frac{f}{k} \right)^2 + 3 \left(\frac{g}{k} \right)^2 = (\lambda^2 + 3\mu^2)(r^2 + 3s^2)^3,$$

an equation which can be solved in rational terms for all values of λ, μ, r, s ; and consequently the product of the squares of the differences of the roots may be a square, and at the same time the roots themselves may be irrational⁹.

⁹Thus then it appears that the total rationality of the roots of the equation $x^3 - qx - r = 0$ may be determined by a direct method without having recourse to the method of divisors to determine the roots themselves; the two conditions being that $4q^3 - 27r^2$ shall be a perfect square, and the greatest common measure of q^3 and r^2 a perfect sixth power.

I believe it will be found on inquiry that the equation $x^n - qx + r = 0$ will always have two rational roots if

$$(n-1)^{n-1} \cdot q^n - n^n \cdot r^{n-1}$$

be a complete square, provided that q be prime to r .

Furthermore, viewing the striking analogy of the general nature of the conditions of rationality already obtained, to those which serve to determine the reality of the roots of equations, I am strongly of opinion that a theorem remains to be discovered, which will enable us to pronounce on the existence of integer, as Sturm's theorem on that of possible roots of a complete equation of any degree: the analogy of the two cases fails however in this respect, that while imaginary roots enter an equation in pairs, irrational roots are limited to entering in groups, each containing two or MORE.

19.

AN ACCOUNT OF A DISCOVERY IN THE THEORY OF NUMBERS RELATIVE TO THE EQUATION $Ax^3 + By^3 + Cz^3 = Dxyz$

[Philosophical Magazine, XXXI. (1847), pp. 189–191]

FIRST GENERAL THEOREM OF TRANSFORMATION.

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If in the equation

$$Ax^3 + By^3 + Cz^3 = Dxyz, \quad (1)$$

A and B are equal, or in the ratio of two cube numbers to one another, and if $27ABC - D^3$ (which I shall call the Determinant) is free from all single or square prime positive factors of the form $6n + 1$, but without exclusion of cubic factors of such form, and if A and B are each odd, and C the double or quadruple of an odd number, or if A and B are each even and C odd, then, I say, the given equation may be made to depend upon another of the form

$$A'u^3 + B'v^3 + C'w^3 = D'uvw;$$

where

$$A'B'C' = ABC, \quad D' = D, \quad uvw = \text{some factor of } z.$$

The following are some of the consequences which I deduce from the above theorem. In stating them it will be convenient to use the term Pure Factorial to designate any number into the composition of which no single or square prime positive factor of the form $6n + 1$ enters.

The equations

$$x^3 + y^3 + 2z^3 = Dxyz, \quad x^3 + y^3 + 4z^3 = Dxyz, \quad 2x^3 + 2y^3 + z^3 = Dxyz,$$

are insoluble in integer numbers, provided that the Determinant in each case is a Pure Factorial.

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The equation

$$x^3 + y^3 + Az^3 = 9Bxyz$$

is insoluble in integer numbers, provided that the Determinant, for which in this case we may substitute $A - 27B^3$, is a pure factorial whenever A is of the form $9n \pm 1$, and equal to $2p^{3i \pm 1}$ or $4p^{3i \pm 1}$, p being any prime number whatever.

I wish however to limit my assertion as to the insolubility of the equations above given. The theorem from which this conclusion is deduced does not preclude the possibility of two of the three quantities x, y, z being taken positive or negative *units*, either in the given equation itself or in one or the other of those into which it may admit of being transformed. Should such values of two

of the variables afford a particular solution, then instead of affirming that the equations are insoluble, I should affirm that the *general solution* can be obtained by equations in finite differences¹⁰.

SECOND GENERAL THEOREM OF TRANSFORMATION.

The equation

$$f^3x^3 + g^3y^3 + h^3z^3 = Kxyz \quad (2)$$

may always be made to depend upon an equation of the form

$$Au^3 + Bv^3 + Cw^3 = Duvw,$$

where

$$ABC = R^3 - S^3, \quad D = 3R;$$

and $uvw =$ some factor of $fx + gy + hz$.

$$R \text{ representing } K + 6fgh, \quad S \quad K - 3fgh.$$

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I have not leisure to show the consequences of this theorem of transformation in connexion with the one first given, but shall content myself with a single numerical example of its applications:

$$x^3 + y^3 + z^3 = -6xyz$$

may be made to depend on the equation

$$u^3 + v^3 + w^3 = 0,$$

and is therefore insoluble.

It is moreover apparent that the Determinant of equation (2) transformed is in general $-27R^3$, and is therefore always a Pure Factorial, and consequently the equation

$$f^3x^3 + g^3y^3 + h^3z^3 = Kxyz$$

will be itself insoluble, being convertible into an insoluble form, provided that $K + 6fgh$ is divisible by 9, and provided further that $(K + 6fgh)^3 - (K - 3fgh)^3$

¹⁰Take for instance the equation $x^3 + y^3 + 2z^3 = 3xyz$. The Determinant 27.25 is a Pure Factorial: consequently if the solution be possible, since in this case the transformed must be identical with the given equation, this latter must be capable of being satisfied by making x and y positive or negative units. Upon trial we find that $x = 1, y = 1, z = 2$ will satisfy the equation. I believe, but have not fully gone through the work of verification, that these are the only possible values (prime to one another) which will satisfy the equation. Should they not be so, my method will infallibly enable me to discover and to give the law for the formation of all the others.

Here, then, under any circumstances, is an example, the first on record, of the complete resolution of a numerical equation of the third degree between three variables.

belongs to the form m^3Q , where Q is of the form $9n \pm 1$, and also of one or the other of the two forms $2p^{3i \pm 1}, 4p^{3i \pm 1}$, p being any prime number whatever.

Pressing avocations prevent me from entering into further developments or simplifications at this present time.

It remains for me to state my reasons for putting forward these discoveries in so imperfect a shape. They occurred to me in the course of a rapid tour on the continent, and the results were communicated by me to my illustrious friend M. Sturm in Paris, who kindly undertook to make them known on my part to the Institute.

Unfortunately, in the heat of invention I got confused about the law of oddness and evenness, to which the coefficients of the given equation are in the first theorem *generally* (in order for the successful application of my method as far as it is yet developed) required to be subject. I stated this law erroneously, and consequently drew erroneous conclusions from my Theorems of Transformation, which I am very anxious to seize the earliest opportunity of correcting. I venture to flatter myself that as opening out a new field in connexion with Fermat's renowned Last Theorem, and as breaking ground in the solution of equations of the third degree, these results will be generally allowed to constitute an important and substantial accession to our knowledge of the Theory of Numbers.

20.

ON THE EQUATION IN NUMBERS $Ax^3 + By^3 + Cz^3 = Dxyz$, AND ITS ASSOCIATE SYSTEM OF EQUATIONS

[*Philosophical Magazine*, XXXI. (1847), pp. 293–296]

p. 110

In the last Number of this *Magazine* I gave an account of a remarkable transformation to which the equation

$$Ax^3 + By^3 + Cz^3 = Dxyz$$

is subject when certain conditions between the coefficients A, B, C, D are satisfied; which conditions I shall begin by expressing with more generality and precision than I was enabled to do in my former communication.

1. Two of the quantities A, B, C are to be to one another in the ratio of two cubes.

2. $27ABC - D^3$ must contain no positive prime factor whatever of the form $6n + 1$. I erred in my former communication in not excluding cubic factors of this form.

3. If 2^m is the highest power of 2 which enters into ABC , and 2^n the highest power of 2 which enters into D , then either m must be of the form $3n \pm 1$, or if not, then m must be greater than $3n$.

These three conditions being satisfied, the given equation can always be transformed into another,

$$A'u^3 + B'v^3 + C'w^3 = D'uvw,$$

where

$$A'B'C' = ABC, \quad D' = D, \quad uvw = \text{a factor of } z.$$

The consequence of this is, as stated in my former paper, that wherever A, B, C, D , besides satisfying the conditions above stated, are taken so as likewise to satisfy the condition,—firstly, of ABC being equal to $2^{3m \pm 1}$, or secondly, of ABC being equal to $2^{3m \pm 1} \cdot p^{3n \pm 1}$, provided in the second case that ABC is of the form $9m \pm 1$, and that D is divisible by 9, p being in both cases a prime, then the given equation will be *generally* insoluble.

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And I am now enabled to add that the *only* solution of which it will in any case admit, is the solitary one found by making two of the terms Ax^3, By^3, Cz^3 equal to one another; so that, for instance, if the given equation should be of the form

$$x^3 + y^3 + ABCz^3 = Dxyz,$$

then the above conditions being satisfied, the one solitary solution of which the equation can possibly admit, is $x = 1, y = 1$,

$$Az^3 - Dz + 2 = 0,$$

which may or may not have possible roots. I call this a *solitary* or singular solution, because it exists alone and no other solution can be deduced from it; whereas in general I shall show that any one solution of the equation

$$Ax^3 + By^3 + Cz^3 = Dxyz$$

can be made to furnish an infinity of other solutions independent of the one supposed given, that is, not reducible thereto by expelling a common factor from the new system of values of x, y, z deduced from the given system.

The following is the Theorem of Derivation in question:

Let

$$A\alpha^3 + B\beta^3 + C\gamma^3 = D\alpha\beta\gamma.$$

Then if we write

$$F = A\alpha^3, \quad G = B\beta^3, \quad H = C\gamma^3,$$

and make

$$x = F^2G + G^2H + H^2F - 3FGH, \quad y = FG^2 + GH^2 + HF^2 - 3FGH,$$

$$z = \frac{1}{D}\{F^3 + G^3 + H^3 - 3FGH\},$$

or

$$= \alpha\beta\gamma\{F^2 + G^2 + H^2 - FG - FH - GH\},$$

we shall have

$$x^3 + y^3 + ABCz^3 = Dxyz.$$

I am hence enabled to show that whenever $x^3 + y^3 + Az^3 = Dxyz$ is insoluble, there will be a whole family of allied equations equally insoluble. For instance, because $x^3 + y^3 + z^3 = 0$ is insoluble in integer numbers, I know likewise that

$$x^6 + y^6 + z^6 = x^3y^3 + x^3z^3 + y^3z^3$$

$$x^6 + y^6 + z^6 = x^3y^3 + x^3z^3 - 2y^3z^3$$

are each equally insoluble.

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In fact

$$\begin{aligned} & (x^3 + y^3 + z^3) \times (x^6 + y^6 + z^6 - x^3y^3 - x^3z^3 - y^3z^3) \\ & \times (x^6 + y^6 + z^6 - x^3y^3 - x^3z^3 + 2y^3z^3) \\ & \times (x^6 + y^6 + z^6 - y^3z^3 - y^3x^3 + 2x^3z^3) \\ & \times (x^6 + y^6 + z^6 - x^3z^3 - z^3y^3 + 2y^3x^3) \\ & = u^3 + v^3 + w^3, \end{aligned}$$

where u, v, w are rational integral functions of x, y, z .

Hence each of the factors must be incapable of becoming zero¹¹.

As a particular instance of my general theory of transformation and elevation, take the equation

$$x^3 + y^3 + 2z^3 = Mxyz.$$

Then, with the exception of the singular or solitary solution $x = 1, y = 1$, of which I take no account, I am able to affirm that for all values of M between 7 and -6 , both inclusive, with the exception of $M = -2$, the equation is insoluble in integer numbers.

Take now the equation where $M = -2$, namely

$$x^3 + y^3 + 2z^3 + 2xyz = 0.$$

One particular solution of this is

$$x = 1, \quad y = -1, \quad z = 1.$$

Another, which I shall call the second¹², is

$$x = 1, \quad y = 3, \quad z = -2.$$

From the first solution I can deduce in succession the following:

$$\begin{array}{lll} x = 11, & y = 5, & z = -7, \\ x = -793269121, & y = 1179490001, & z = -1189735855, \\ \&c. & \&c. & \&c. \end{array}$$

From the second,

$$\begin{array}{lll} x = -10085, & y = 8921, & z = -8442, \\ x = \&c. & y = \&c. & z = \&c. \end{array}$$

As another example, take the equation

$$x^3 + y^3 + 6z^3 = 6xyz.$$

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One solution of the transformed equation

$$u^3 + 2v^3 + 3w^3 = 6uvw$$

is evidently

$$u = 1, \quad v = 1, \quad w = 1.$$

¹¹It is however sufficiently evident from their intrinsic form, which may be reduced to $\frac{1}{4}(M^2 + 3N^2)$, that this impossibility exists for all the factors except the first.

¹²See Postscript.

Hence I can deduce an infinite series of solutions of the given equation, of which the first in order of ascent will be

$$x = 5, \quad y = 7, \quad z = 3.$$

Again, the lowest possible solution in integers of the equation

$$x^3 + y^3 + 6z^3 = 0$$

will be

$$x = 17, \quad y = 37, \quad z = -21.$$

The equation

$$x^3 + y^3 + 9z^3 = 0$$

admits of the solutions

$$\begin{aligned} x &= 1, & y &= 2, & z &= -1, \\ x &= -271, & y &= 919, & z &= -438. \end{aligned}$$

I trust that my readers will do me the justice to believe that I am in possession of a strict demonstration of all that has been here advanced without proof. Certain of the writer's friends on the continent have, in their comments upon one of his former papers which appeared in this *Magazine*, complimented his powers of divination at the expense of his judgment, in rather gratuitously assuming that the author of the Theory of Elimination was unprovided with the demonstrations, which he was too inert or too beset with worldly cares and distractions to present to the public in a sufficiently digested form. The proof of whatever has been here advanced exists not merely as a conception of the author's mind, but fairly drawn out in writing, and in a form fit for publication.

P.S. It must not be supposed that the two primary or basic solutions above given of the equation

$$x^3 + y^3 + 2z^3 + 2xyz = 0,$$

namely,

$$\begin{aligned} x &= 1, & y &= -1, & z &= 1, \\ x &= 1, & y &= 3, & z &= -2, \end{aligned}$$

are independent of one another. The second may be derived from the first, as I shall show in a future communication. In fact there exist *three* independent processes, by combining which together, one particular solution may be made to give rise to an infinite series of infinite series of infinite series of correlated solutions, which it may possibly be discovered contain between them the *general* complete solution of the equation

$$x^3 + y^3 + Az^3 = Dxyz.$$

21.

ON THE GENERAL SOLUTION (IN CERTAIN CASES) OF THE EQUATION

$$x^3 + y^3 + Az^3 = Mxyz, \text{ \&C.}$$

[*Philosophical Magazine*, XXXI. (1847), pp. 467-471]

p. 114

I shall restrict the enunciation of the proposition I am about to advance to much narrower limits than I believe are necessary to the truth, with a view to avoid making any statement which I may hereafter have occasion to modify. Let us then suppose in the equation

$$x^3 + y^3 + Az^3 = Mxyz$$

that A is a *prime* number, and that $27A - M^3$ is *positive*, but exempt from positive prime factors of the form $6i + 1$. Then I say, and have succeeded in demonstrating, that all the possible solutions in integer numbers of the given equation may be obtained by explicit processes from one particular solution or system of values of x, y, z , which may be called the Primitive system.

This system of roots or of values of x, y, z is that system in which the value of the greatest of the three terms $x, y, A^{\frac{1}{3}} \cdot z$ (which may be called the Dominant) is the least possible of all such dominants. I believe that in general the system of the least Dominant is identical with the system of the least Content, meaning by the latter term the product of the three terms out of which the Dominant is elected. I proceed to show the law of derivation.

To express this simply, I must premise that I shall have to employ such an expression as $S' = \phi(S)$ to indicate, not that a certain quantity, S' , is a function of S , but that a certain system of quantities disconnected from one another, denoted by S' , are severally functions of a certain other system of quantities denoted by S ; and, as usual, I shall denote $\phi\phi S$ by $\phi^2 S$, $\phi\phi^2 S$ by $\phi^3 S$, and so forth.

Let now P be the Primitive system of solution of the equation

$$x^3 + y^3 + Az^3 = Mxyz,$$

P denoting a certain system of values of and written in the order of the letters x, y, z , which may always be found by a limited number of trials (provided that the equation admits of any solution).

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That this is the case is obvious, since we have only to give the Dominant every possible value from the integer next greatest to $A^{\frac{1}{3}}$ upwards, and combine the values of x^3, y^3, Az^3 so that none shall ever exceed at each step the cube of such dominant, and we must at last, if there *exist any solution*, arrive at the System of the Least Dominant.

Now, every system of solution is of one or the other of two characters. Either x and y must be odd and z even, or x and y must be one odd and the other even and z odd. That all three should be odd is inconsistent with the given conditions as to A being odd and M even; and if all three were even, by driving out the common factor we should revert to one or the other of the foregoing cases.

The systems of solution where z is even may be termed Reducible, those where z is odd Irreducible. Let ϕ denote a certain symbol of transformation hereafter to be explained.

Then the Reducible systems of the first order may be expressed by

$$\phi P, \phi^2 P, \phi^3 P, \text{ ad infinitum};$$

or in general by $\phi^{n_1} P$, n_1 being absolutely arbitrary. I will anticipate by stating that the function ϕ involves no *variable* constants; that is to say, $\phi(S)$ may be found explicitly from S without any reference to the particular equation to which S belongs. Let now ψ denote another symbol of transformation, also hereafter to be defined, and differing from ϕ insofar as it does involve as *constants* the three values of x, y, z contained in P : then the general representations of Irreducible systems of the first order will be denoted by $\psi\phi^{n_1} P$.

It is proper to state here that the symbol ψ is ambiguous; and $\psi\phi^{n_1} P$, when P and n_1 are given, will have two values, according to the way in which the terms represented by P are compared with x, y, z in the given equation

$$x^3 + y^3 + Az^3 = Mxyz;$$

for it is obvious that if $x = a, y = b, z = c$ satisfies the equation, so likewise will

$$x = b, \quad y = a, \quad z = c.$$

Each however of these values of $\psi\phi^{n_1} P$ gives a solution of the kind above designated.

Proceeding in like manner as before, the Reducible system of the second order may be designated by $\phi^{n_2} \cdot \psi\phi^{n_1} \cdot P$, the Irreducible by $\psi\phi^{n_2} \cdot \psi\phi^{n_1} \cdot P$; and in general *every possible* system of values of x, y, z satisfying the proposed equation, in which z is even, is comprised under the form

$$\phi^{n_r} \cdot \psi\phi^{n_{r-1}} \cdot \psi \dots \psi\phi^{n_1} \cdot P;$$

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and every possible system of such values, in which z is odd, is comprised under the form

$$\psi\phi^{n_r} \cdot \psi\phi^{n_{r-1}} \cdot \psi \dots \psi\phi^{n_1} \cdot P :$$

the quantities $n_1, n_2 \dots n_r$ being of course all independent of one another, and unlimited in number and value.

Thus then we may be said to have the general solution of the given equation in the same sense as an arbitrary sum of terms, each of a certain form, is in certain cases accepted as the complete solution of a partial differential equation.

As regards the value of the symbols ψ and ϕ , ϕ indicates the process by which a, b, c becomes transformed into α, β, γ , the relations between the two sets of elements being contained in the following equations:

$$\begin{aligned} a' &= a^3, & b' &= b^3, & c' &= Ac^3, \\ \alpha &= a'^2b' + b'^2c' + c'^2a' - 3a'b'c', \\ \beta &= a'b'^2 + b'c'^2 + c'a'^2 - 3a'b'c', \\ \gamma &= abc\{a'^2 + b'^2 + c'^2 - a'b' - a'c' - b'c'\}. \end{aligned}$$

Next, as to the effect of the Duplex symbol ψ . Let e, g, ι be the elements of the Primitive system P : ι being the value of z and e, g of x and y taken in either mode of combination, each with each, which satisfy the proposed equation

$$x^3 + y^3 + Az^3 = Mxyz.$$

Let l, m, n represent any system S ,

$$\lambda, \mu, \nu \text{ represent any system } \psi(S),$$

ψS has two values, which we may denote by $\psi'S, '\psi S$ respectively, and accentuating the elements λ, μ, ν accordingly to correspond, we shall have

$$\begin{aligned} \lambda' &= 3gm(gl - em) + 3A\iota n(\iota l - en) - M(gl^2 - e^2lm), \\ \mu' &= 3A\iota n(\iota m - gl) + 3el(em - gl) - M(e\iota m^2 - g^2lm), \\ \nu' &= 3el(en - \iota l) + 3gm(gn - \iota m) - M(egn^2 - \iota^2lm) : \end{aligned}$$

we have then

$$\psi'S = \lambda', \mu', \nu',$$

and in like manner

$$' \psi S = \lambda, \mu, \nu,$$

$' \psi S$ being derived from $\psi'S$ by the mere interchange of e and g one with the other.

p. 117

I have stated that every possible solution of the proposed equation comes under one or the other of the orders, infinite in number and infinite to the power of infinity in variety of degree, above given: this is not strictly true, unless we understand that all systems of solution are considered to be equivalent which differ only in a multiplier common to all three terms of each; that is to say, which may be rendered identical by the expulsion of a common factor. So that $m\alpha, m\beta, m\gamma$ as a system is treated as identical with α, β, γ , which of course

substantially it is; and it should be remarked that there is nothing to prevent the operations denoted by ϕ and ψ introducing a common factor into the systems which they serve to generate, and the latter in particular will have a strong tendency so to do.

I believe that this theorem may be extended with scarcely any modification to the case where A , instead of being a prime, is any power of the same, and to suppositions still more general. I believe also that, subject to certain very limited restrictions, the theorem may prove to apply to the case where the determinant $27A - M^3$ becomes negative.

The peculiarity of this case which distinguishes it from the former, is that it admits of all the three variables x, y, z in the equation

$$x^3 + y^3 + Az^3 = Mxyz$$

having the same sign, which is impossible when the determinant is positive; or in other words, the curve of the third degree represented by the equation

$$Y^3 + X^3 + 1 = \frac{M}{A^{\frac{1}{3}}}XY$$

(in which I call the coefficient of XY the characteristic), which, as long as the quantity last named is less than 3, is a single continuous curve extending on both sides to infinity, as soon as the characteristic becomes equal to 3 assumes to itself an isolated point, the germ of an oval or closed branch, which continues to swell out (always lying apart from the infinite branch) as the characteristic continues indefinitely to increase.

I ought not to omit to call attention to the fact that the theorem above detailed is always applicable to the case of the equation

$$x^3 + y^3 + Az^3 = 0,$$

when A is *any* power of a prime number *not* of the form $6i + 1$; in other words, the above always belongs to the class of equations having Monogenous solutions, which for the sake of brevity may be termed themselves Monogenous Equations¹³. p. 118

On the probable existence of such a class of equations I hazarded a conjecture at the conclusion of my last communication to this *Magazine*. As I hope shortly to bring out a paper on this subject in a more complete form, I shall content myself at this time with merely stating a theorem of much importance to the completion of the theory of insoluble and of Monogenous equations of the third degree; to wit, that the equation in integers

$$a(x^3 + y^3 + z^3) + c(x^2y + y^2z + z^2x + xy^2 + yz^2 + zx^2) + exyz = 0$$

¹³Thus the equation $x^3 + y^3 + 9z^3 = 0$ alluded to by Legendre is Monogenous, and the Primitive system of solution is $x = 1, y = 2, z = -1$, from which every other possible solution in Integers may be deduced.

may always be transformed so as to depend upon the equation

$$fu^3 + gv^3 + hw^3 = (6a - e)uvw,$$

wherein

$$fgh = ae^2 - (c^2 + 3a^2)e + 9a^2 - 3ac^2 - 2c^3.$$

By means of the above theorem, among other and more remarkable consequences, we are enabled to give a theory of the irresoluble and monogenous cases of the equation

$$x^3 + y^3 + m^3z^3 = Mxyz,$$

when m is some power of 2, or of certain other numbers.